Dopamine, via its receptors, plays a vital role in the maintenance of blood pressure by modulating renal sodium transport. However, the role of the D_4 dopamine receptor (D_4 receptor) in renal proximal tubules (PRTs) is still unclear. We hypothesized that activation of D_4 receptor directly inhibits the activity of the Na^+-K^+-ATPase (NKA) in RPT cells.